

Multimedica Scanner Appliance Installation Guide

Audience: Technical installer **Applies to:** New or freshly reimaged Raspberry Pi scanner appliances **Qualified baseline:** Raspberry Pi 4 Model B, Raspberry Pi OS Lite 64-bit, Debian 13 Trixie **Last validated:** August 2026

1. Purpose

This guide describes the supported procedure for turning a clean Raspberry Pi into a commissioned Multimedica scanner with a validated production release.

Use the Windows provisioning script for installation and verification. Do not clone the repository onto the Pi, copy individual runtime files manually, or edit files under `/opt/multimedica-scanner` as part of ordinary installation.

2. Where the scanner appliance fits

The scanner appliance is one component of the complete **Alfarero Clinic (Multimédica)** system. It is not a stand-alone application. The appliance reads configuration QR codes during commissioning and patient workflow barcodes during everyday clinic operation, while the main Multimédica application supplies the configuration and patient workflow information.

Authorized users generate the required scanner configuration QR codes in the main Multimédica application at <https://multimedica.org>. Those codes configure the appliance's Wi-Fi, station identity, and cloud connection. The installer should obtain the approved QR codes from an authorized Multimédica user; the installer should not compose or modify their contents manually.

Patient workflow tickets are printed from the **Anfitrión** page of the Multimédica application. The clinic must use one of these supported printing arrangements:

1. **Page printer:** Print from Windows to a compatible conventional printer.
2. **Thermal ticket printer:** Install and configure the separate `local-print-server` application on a Windows computer that can communicate with the thermal printer and is reachable over the clinic network from the users' devices.

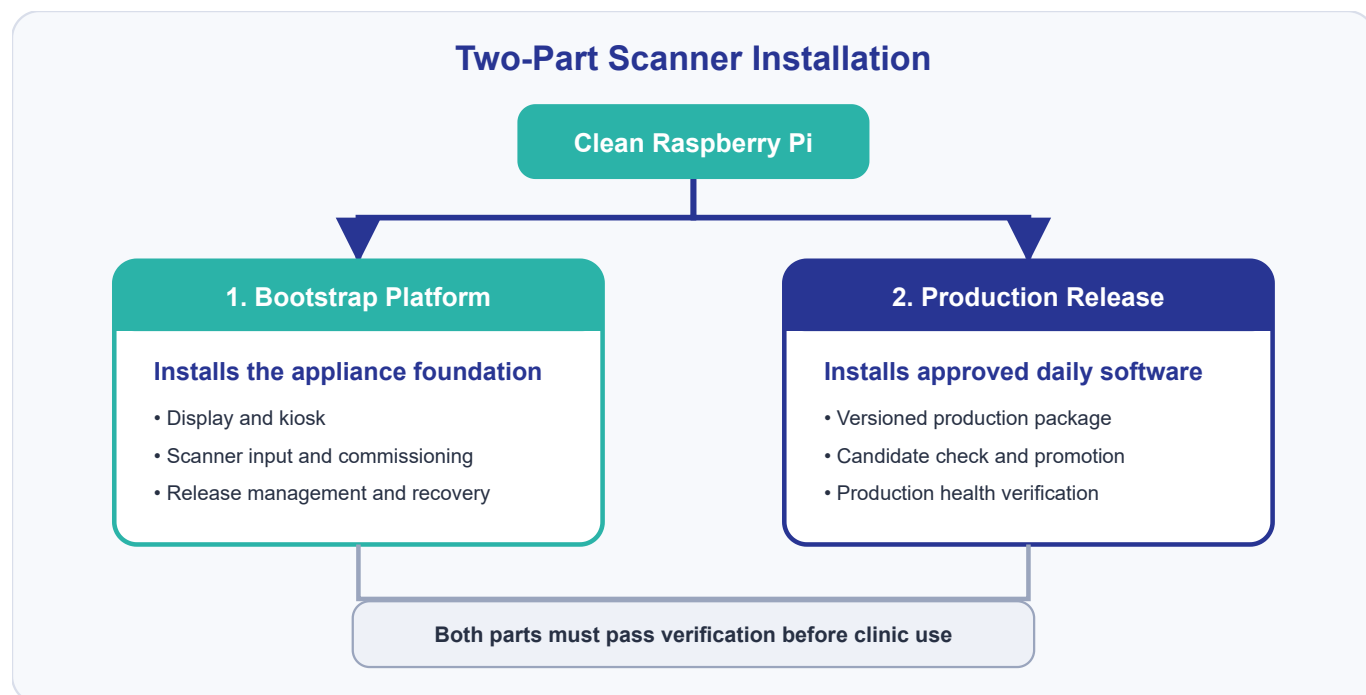
The thermal printing path is a separate installation responsibility and is not installed by this scanner-appliance procedure. Before commissioning a clinic that uses thermal tickets, confirm that the Windows print-server computer, `local-print-server`, thermal printer, and relevant network access are available and working. Follow the separate `local-print-server` [installation instructions](#).

2.1 What this procedure installs on the appliance

The completed appliance contains two software layers:

1. **Bootstrap layer:** scanner input, QR commissioning, display, kiosk, release management, and recovery.
2. **Production release:** versioned everyday scan processing and clinic-cloud communication.

These layers are installed and accepted separately. A completed bootstrap installation does not by itself mean production is installed.



The bootstrap path establishes and commissions the appliance platform. The production path installs a specific approved release only after the bootstrap platform has passed verification.

For a developer-oriented explanation of how these layers work together—including commissioning, release promotion, rollback, recovery, gates, and markers—see [Scanner Installation Theory of Operation](#). Reading it is optional for an installer following this procedure, but recommended for anyone maintaining or troubleshooting the system.

3. Command conventions

This guide uses these terms consistently:

- **Pi:** the Raspberry Pi 4 computer inside the appliance.
- **Scanner:** the USB barcode scanner connected to the Pi.
- **Scanner appliance:** the complete assembly: Pi, display, USB scanner, enclosure, storage, and installed software.
- **Provisioning workstation:** the Windows computer from which the installer runs the installation commands.

Label	Meaning
Windows	Run in Windows PowerShell from the repository root
Scanner	Scan a printed QR with the appliance barcode scanner
Physical	Observe or manipulate the Pi, display, cables, or power

PowerShell continuation commands use the backtick character at the end of each continued line. Do not add characters after the backtick.

The repository root is the directory containing:

```
provision-scanner.ps1
package.json
bootstrap/
production/
tests/
```

Installation worksheet

Use this worksheet to record the non-secret values that will be reused throughout this procedure. Print it or copy it onto paper, and keep it available while imaging and provisioning the Pi. Record the Pi password only in the approved password manager or other secure credential record—not in this worksheet.

MULTIMÉDICA SCANNER INSTALLATION WORKSHEET

Pi hostname:

Pi username (required):

multimedica_edge

PiHost value (username@hostname):

Pi password stored in approved secure location: ☐ Yes

At the beginning of the provisioning session, copy and paste the complete block below into Windows PowerShell. The commands will ask for the worksheet values and store them for use by later commands.

Windows PowerShell:

```
$piUser = Read-Host "Enter the Pi username recorded during imaging"
$piHostname = Read-Host "Enter the Pi hostname recorded during imaging"
$piHost = "$piUser@$piHostname"
$piHost
```

Confirm that the displayed `$piHost` combines the correct username and hostname. The remaining PowerShell examples use `$piHost` and `$piHostname`; set them again after opening a new PowerShell window.

4. Stop conditions

Stop the procedure and investigate if any of the following occurs:

- the provisioning script reports **RESULT: FAIL**;
- SSH reports an unexpected host-key change and the Pi was not deliberately reimaged;
- a password is displayed in clear text or appears in captured output;
- a required QR is rejected;
- the scanner USB device or reader is not detected;
- the physical display does not show **CANDIDATO** during release installation;
- production health does not pass;
- the final clinic display or real-patient scan is incorrect.

Do not continue merely because a later step might appear to work. Preserve **provisioning-result.json** and the console output for diagnosis.

5. Required hardware

- Raspberry Pi 4 Model B
- 32 GB or larger high-quality microSD card
- Qualified HDMI portrait display
- HDMI jumper or cable required by the display
- Display standoffs and mounting hardware
- Compatible USB barcode/QR scanner that presents scans as keyboard-style input
- Raspberry Pi USB-C power supply
- Right-angle USB-C cable when required by the enclosure
- Ethernet cable or Wi-Fi network access
- Completed scanner enclosure and suitable fasteners

Enclosure fabrication option

If a completed enclosure is not available, the installer may fabricate one from the two 3D-printing files included in the repository's **docs** folder:

- **Enclosure Top.stl**
- **Enclosure Bottom.stl**

The files may be printed locally or sent to a commercial 3D-printing service. **PET-G is the recommended filament** because it provides a useful balance of strength, heat resistance, and printability, but PET-G is not mandatory. Another suitable material may be used when the finished enclosure is sufficiently rigid, dimensionally stable, and appropriate for the clinic environment.

Qualified reference hardware and image details are recorded in [Bootstrap Architecture](#).

6. Hardware assembly

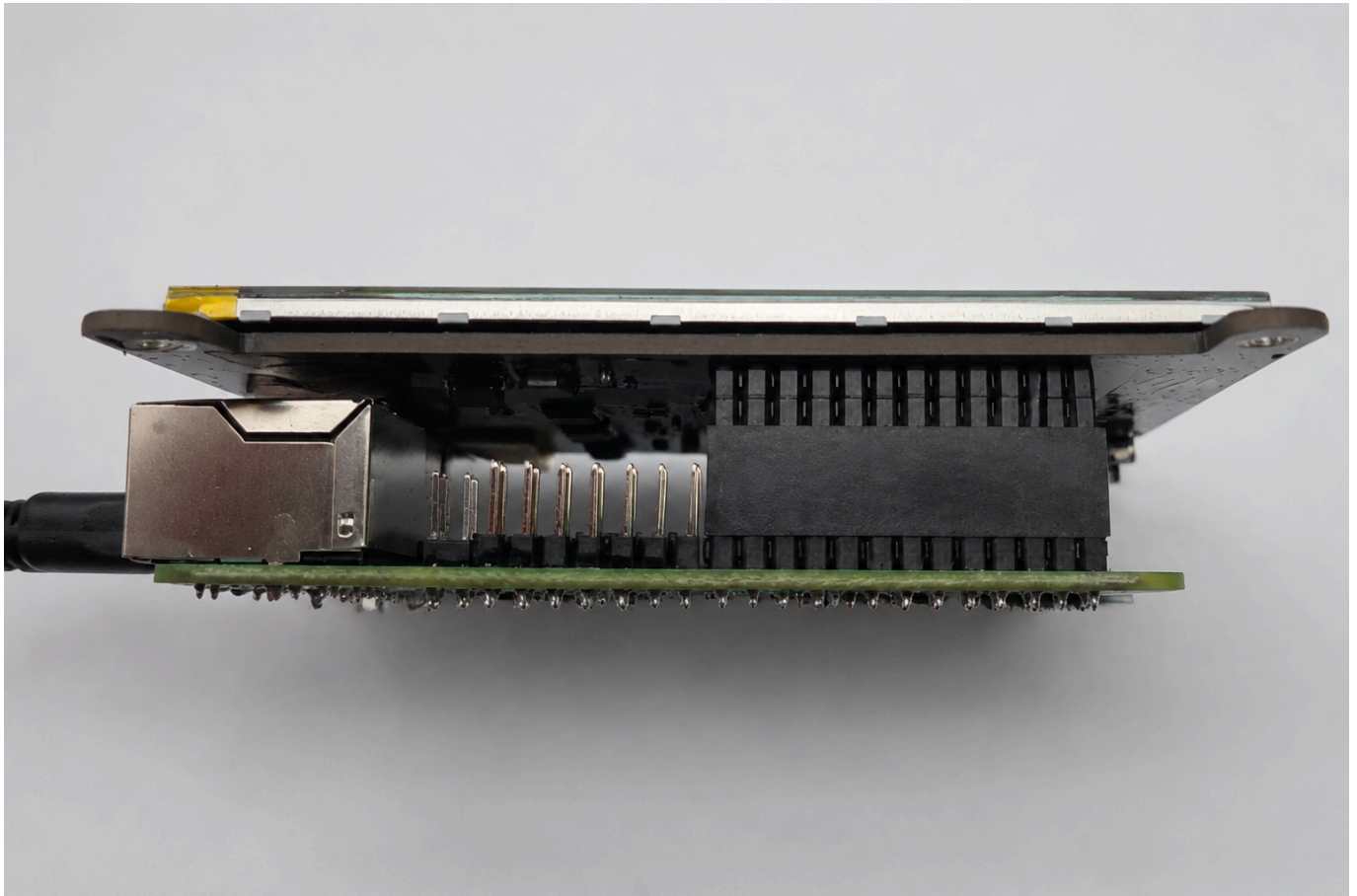
Perform assembly with power disconnected.

At this stage, the microSD card has probably not yet been imaged with Raspberry Pi OS. Assemble and test-fit the display, Pi, scanner, and cables, but do **not** permanently place the electronics into the enclosure yet. After

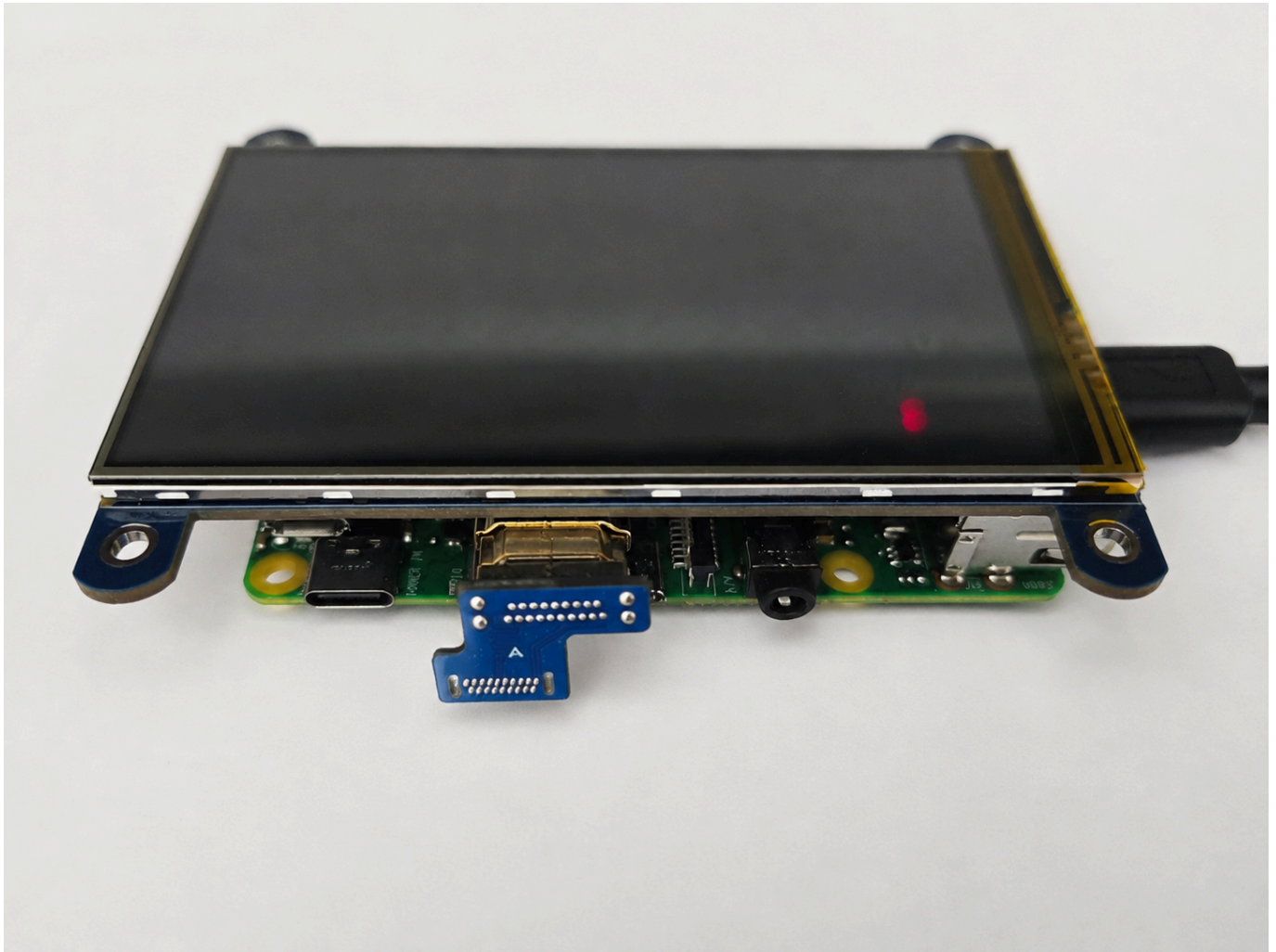
imaging, the microSD card must be inserted into the Pi. Final enclosure installation, cable routing, and cable dressing are completed after Section 9.15 and before first boot.

6.1 Mount the display

1. Align the display connector with the end of the Raspberry Pi GPIO header exactly as shown. The ends of the two connectors must line up; unused GPIO pins remain visible at the opposite side.
2. Confirm every receiving position is over the correct pin before applying pressure. Never force the display onto a misaligned header.



3. Install the HDMI jumper between the display and the Pi micro-HDMI port in the orientation shown.
4. Confirm both ends of the jumper are fully seated and that the jumper is not twisted or under tension.



5. Install the required standoffs and screws.

The completed assembly must be rigid, with no bent GPIO pins or stressed HDMI connectors.

6.2 Connect the scanner

Connect the compatible USB barcode/QR scanner to a Raspberry Pi USB port. Do not assume the Linux event-device number; the bootstrap reader discovers the device dynamically.

6.3 Plan cable routing

The appliance operates in portrait orientation. Test-fit the scanner, network, and other external cables through the intended enclosure exits. Use a right-angle USB-C power cable when necessary to avoid enclosure interference. Leave the electronics accessible so the imaged microSD card can be inserted later.

6.4 Hardware check

- ☐ Display firmly mounted
- ☐ HDMI connection fully seated
- ☐ Scanner connected by USB
- ☐ microSD card available for imaging
- ☐ Network connection available
- ☐ Power cable fits without stress
- ☐ No exposed conductor, pin, or board is shorted by the enclosure

7. Prepare the Windows provisioning workstation

Your Windows computer will be the installation tool used to program and configure the Multimédica scanner. If you have not prepared it previously, complete every step in this section before imaging the Pi. The Windows computer requires internet access while downloading the tools and installation repository.

7.1 Install Git for Windows

Download Git for Windows from <https://git-scm.com/download/win> and run the installer. The standard installation options are suitable unless the organization maintains another approved configuration. Close and reopen Windows PowerShell, then run:

Windows PowerShell:

```
git --version
```

The command must return a version. If PowerShell reports that `git` is not recognized, stop and correct the installation or PATH.

7.2 Install Node.js and npm

Download a supported Node.js LTS release from <https://nodejs.org/en/download>. Install Node.js with npm included. Close and reopen Windows PowerShell, then run:

Windows PowerShell:

```
node --version  
npm --version
```

Both commands must return version numbers.

7.3 Confirm Windows PowerShell

Open **Windows PowerShell** and run:

```
$PSVersionTable.PSVersion
```

Windows PowerShell 5.1 or a later compatible PowerShell is required. Do not paste Bash commands containing syntax such as `<<EOF` into PowerShell.

7.4 Install Raspberry Pi Imager

Download Raspberry Pi Imager from <https://www.raspberrypi.com/software/> and install it. The screenshots in this guide use version 2.0.11.1; later versions may differ slightly while retaining the same required settings.

7.5 Obtain the installation repository

The supported installation repository is <https://github.com/paullmullen/multimedica-scanner>.

For a first checkout, run the following in Windows PowerShell:

Windows PowerShell — first checkout:

```
New-Item -ItemType Directory -Force C:\dev | Out-Null
Set-Location C:\dev
git clone https://github.com/paullmullen/multimedica-scanner.git
Set-Location C:\dev\multimedica-scanner
```

For an existing checkout, first confirm that it contains no uncommitted work.

Windows PowerShell — existing checkout:

```
Set-Location C:\dev\multimedica-scanner
git status --short
```

Expected result: no output at all before the next PowerShell prompt:

```
PS C:\dev\multimedica-scanner> git status --short
PS C:\dev\multimedica-scanner>
```

If any filename or status code appears, the result is not clean. Examples include **M**, **D**, or **??** followed by a filename. Stop and seek qualified help; do not discard another developer's work.

Update the approved branch without creating a merge commit.

Windows PowerShell:

```
git switch main
git pull --ff-only origin main
git status --short
git log -1 --oneline
```

Record the displayed commit in the installation record. **git status --short** should return nothing.

The second **git status --short** has the same expected result: no output before the prompt. Any filename means stop.

Install the repository's exact Node dependencies.

Windows PowerShell:

```
npm ci
```

7.6 Workstation readiness check

- ☐ `git --version` succeeds
- ☐ `node --version` succeeds
- ☐ `npm --version` succeeds
- ☐ Raspberry Pi Imager opens
- ☐ Repository is at the approved `main` commit
- ☐ `git status --short` returns nothing
- ☐ `npm ci` completed successfully
- ☐ PowerShell prompt is at the repository root

8. Prepare configuration and release materials

Your Windows computer is now ready to act as the installation tool. Next, make sure it has all the Multimédica scanner software, configuration information, QR codes, and production-release files needed to complete the installation. Complete this section before installing software on the Pi.

8.1 Confirm the approved repository revision

Use an approved, tested repository revision. Do not perform development changes during a field installation.

From the repository root, confirm the working tree is clean.

Windows PowerShell:

```
git status --short
```

Expected: no output. If any filename appears, stop and resolve it before installation.

For an approved installation package rather than a Git checkout, extract it into a local working directory while preserving its folder structure.

8.2 Unblock the provisioning script when required

Windows may mark software downloaded from the internet as potentially unsafe and prevent it from running. Because this script came from the approved Multimédica repository, remove that Windows block before using it. Only the main provisioning script needs this command.

From **Windows PowerShell** in the repository root, run:

```
Unblock-File .\provision-scanner.ps1
```

8.3 Prepare the installer configuration

The protected installer configuration contains the QR administrator token used to authorize commissioning QRs.

If an approved `multimedica-installer.json` already exists, run:

Windows PowerShell:

```
Test-Path .\multimedica-installer.json
```

Expected:

```
True
```

If it must be created, run:

Windows PowerShell:

```
.\provision-scanner.ps1 -CreateInstallerConfig
```

Follow the hidden token prompt. Keep this file secure. Do not email it, place it in shared storage, include it in screenshots, or distribute it with ordinary installation files.

8.4 Prepare the configuration QRs

During setup, the scanner needs information about its network connection, clinic assignment, and authorized access to the Multimédica cloud software. Because the scanner has no keyboard, the installer supplies this information by scanning QR codes.

An authorized administrator must sign in at <https://multimedica.org>, open the administrative section, and generate or print the QR codes for this scanner. Prepare one of each required code before beginning installation:

- Wi-Fi configuration QR
- Station configuration QR
- Cloud configuration QR
- **Mostrar Identidad del scanner** QR

Printed Wi-Fi and Cloud QRs contain credentials. Control and destroy unneeded copies appropriately.

Keep the **Mostrar Identidad del scanner** QR for the final acceptance check in Section 15.3 and for future troubleshooting. It displays the configured location, room, station, device ID, current IP address, and installed production version without changing the scanner configuration.

8.5 Obtain the production artifact

The final scanner software is supplied as an approved release package in the repository. The release includes:

- a `.tgz` archive containing the production software;
- a `.sha256` checksum sidecar containing the archive's expected digital fingerprint;
- a `.build.json` file recording how the release was built; and
- `approved-production-release.json`, which identifies the version approved for installation.

A SHA-256 hash is a long value calculated from the exact bytes in a file. If a download is incomplete, damaged, or unexpectedly changed, its calculated hash will not match the approved value. The checksum sidecar is a small companion file containing another recorded copy of the expected hash.

The approved repository revision must already contain all four release files:

```
release-output/approved-production-release.json
release-output/multimedica-production-<version>.tgz
release-output/multimedica-production-<version>.tgz.sha256
release-output/multimedica-production-<version>.build.json
```

The repository already identifies the approved production version; you do not need to choose or type a version number. From **Windows PowerShell** in the repository root, run the complete block below. It finds the approved version, confirms that every required release file is present, and verifies the archive's SHA-256 fingerprint before installation:

```
$approvedReleasePath = ".\release-output\approved-production-release.json"
if (-not (Test-Path $approvedReleasePath)) { throw "The approved-release record is
missing from the repository." }

$approvedRelease = Get-Content $approvedReleasePath -Raw | ConvertFrom-Json
$releaseVersion = [string]$approvedRelease.version
$artifactFile = Join-Path ".\release-output" ([string]$approvedRelease.artifact)
$checksumFile = "$artifactFile.sha256"
$buildMetadataFile = Join-Path ".\release-output"
([string]$approvedRelease.build_metadata)

foreach ($requiredFile in @($artifactFile, $checksumFile, $buildMetadataFile)) {
    if (-not (Test-Path $requiredFile)) { throw "Approved release is incomplete:
$requiredFile" }
}

$artifact = (Resolve-Path $artifactFile).Path
$sha = (Get-FileHash $artifact -Algorithm SHA256).Hash.ToLowerInvariant()
$approvedSha = ([string]$approvedRelease.sha256).ToLowerInvariant()
$sidecarSha = ((Get-Content $checksumFile -Raw).Trim() -split '\s+')
[0].ToLowerInvariant()

if ($sha -ne $approvedSha) {
    throw "Production artifact does not match the approved SHA-256."
}
if ($sha -ne $sidecarSha) {
```

```
    throw "Production artifact does not match its checksum sidecar."  
}
```

```
$artifact  
$releaseVersion  
"Calculated SHA-256: $sha"  
"Approved SHA-256:   $approvedSha"  
"Sidecar SHA-256:   $sidecarSha"
```

The three displayed SHA-256 values must be identical. If the approval record or any required release file is absent—or if the command reports that the values do not match—stop. The repository is not a complete, verified installation package.

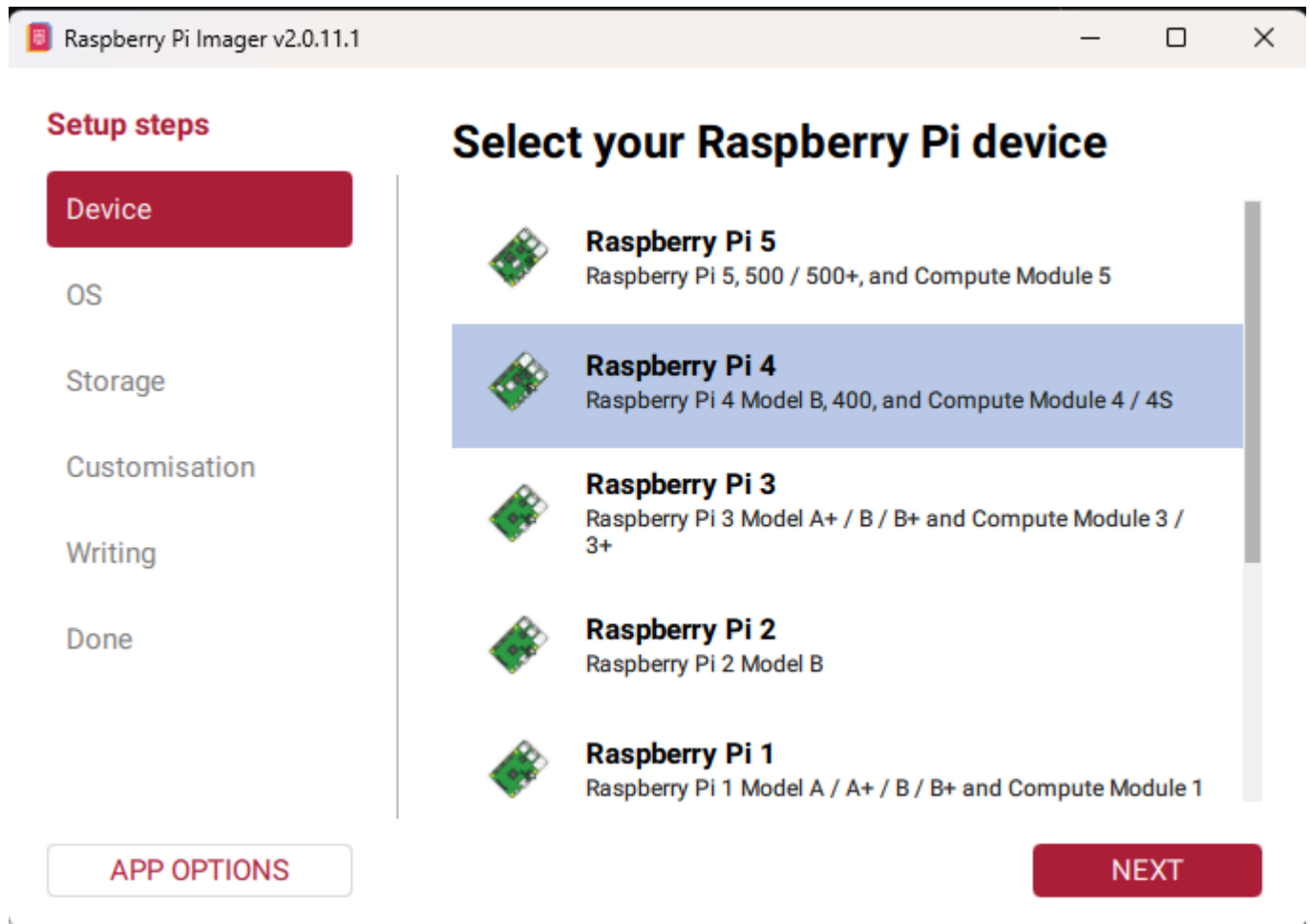
9. Image the Raspberry Pi

9.1 Approved hardware and image

Raspberry Pi 4 Model B is currently the only validated hardware platform. Do not substitute Raspberry Pi 3, Raspberry Pi 5, or another computer without completing platform qualification and physical acceptance testing.

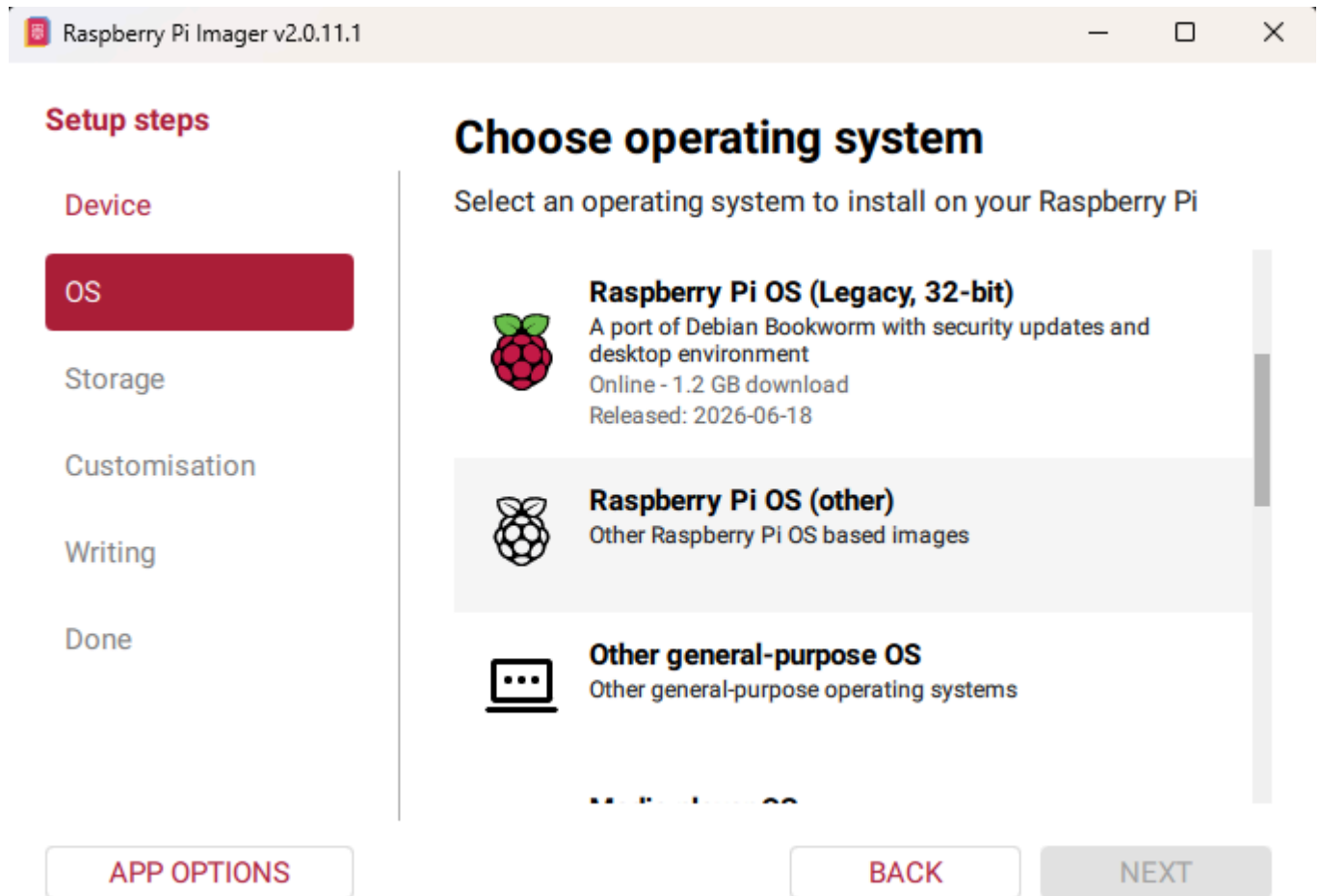
Use the current Debian Trixie-based Raspberry Pi OS Lite 64-bit image offered by Raspberry Pi Imager. Do not substitute Raspberry Pi OS Full/Desktop, a 32-bit image, or an unqualified major OS release.

9.2 Select Raspberry Pi 4



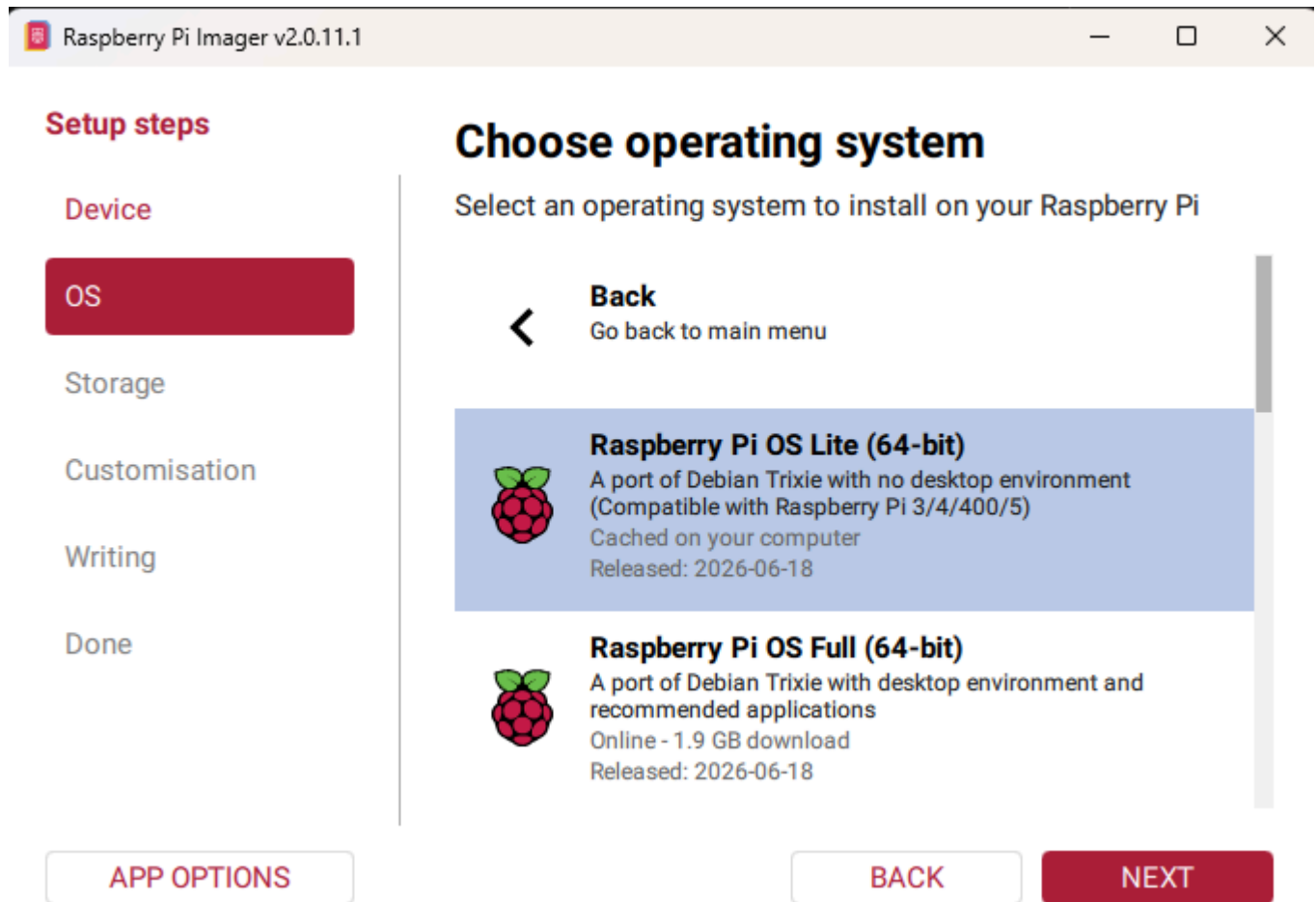
Select **Raspberry Pi 4**, confirm that its row is highlighted, and click **NEXT**.

9.3 Open the alternate Raspberry Pi OS list



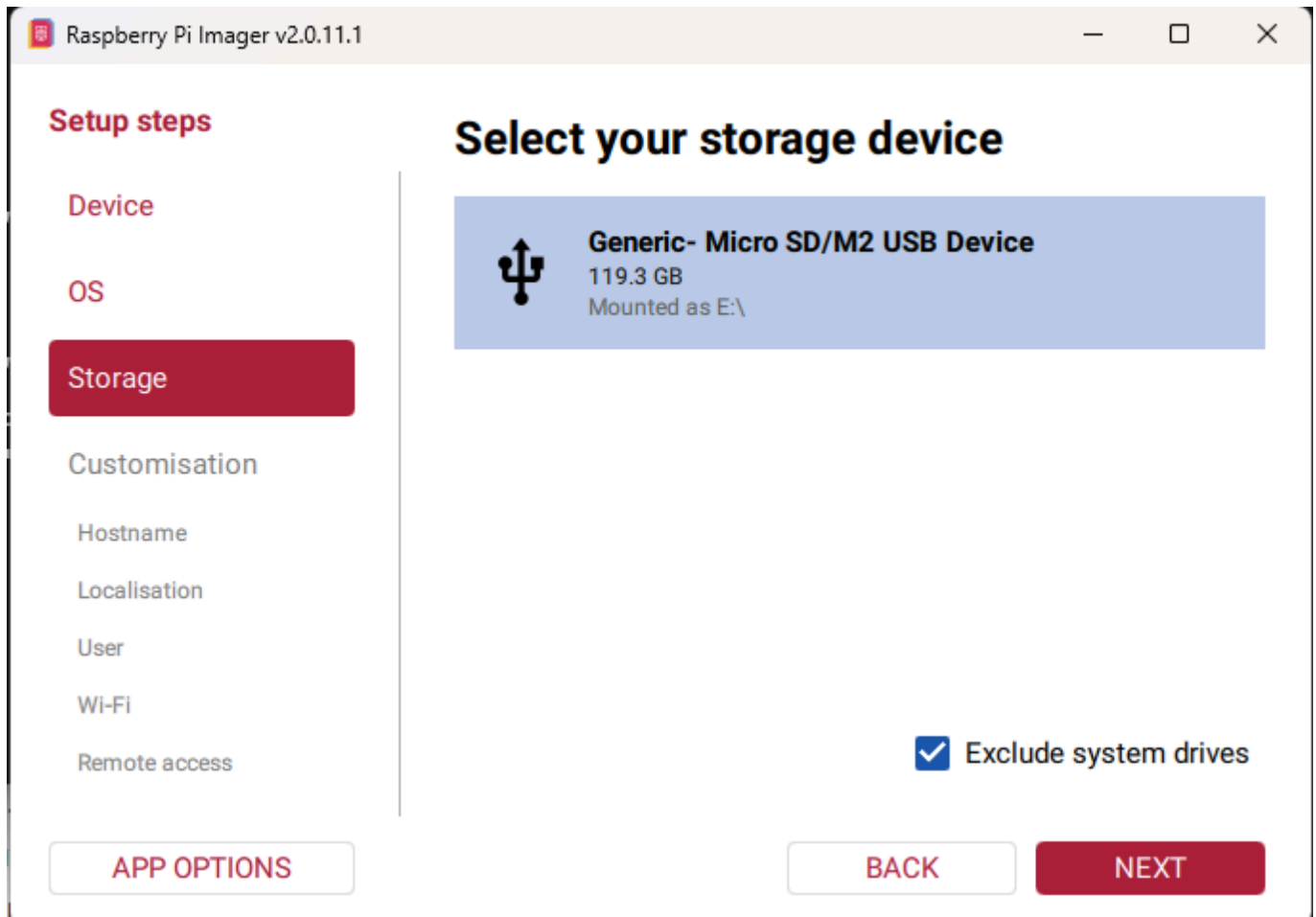
Scroll down and select **Raspberry Pi OS (other)**. Do not select a visible legacy 32-bit image or **Other general-purpose OS**.

9.4 Select Raspberry Pi OS Lite 64-bit



Select **Raspberry Pi OS Lite (64-bit)** and click **NEXT**. The appliance installs its own kiosk environment; do not select Raspberry Pi OS Full.

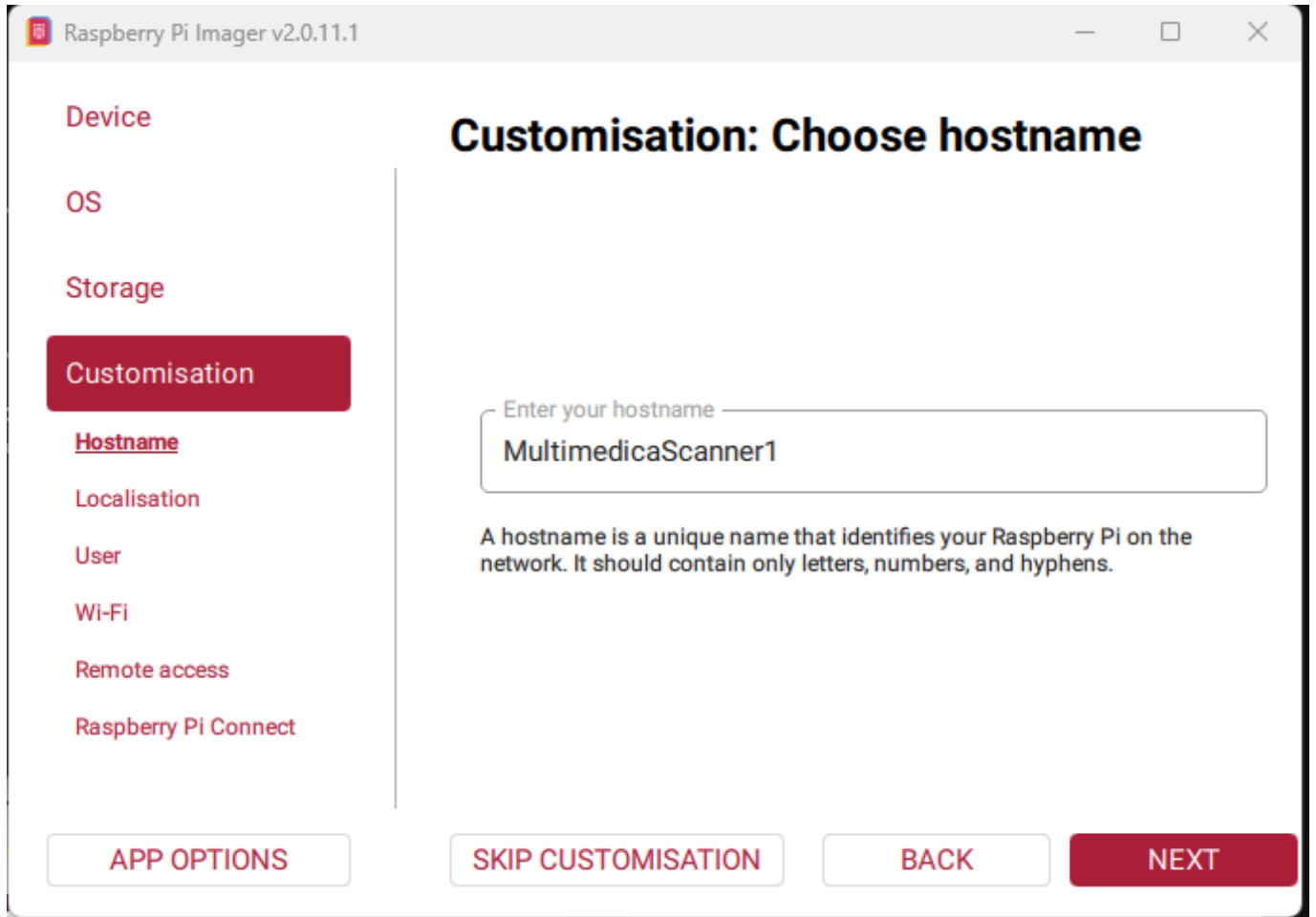
9.5 Select the microSD card



Select the microSD card that will be installed in the scanner appliance, then click **NEXT**. Its manufacturer, capacity, device name, and drive letter may differ from the example.

Warning: Writing the image permanently erases the selected device. Confirm its capacity and Windows drive letter. Keep **Exclude system drives** selected, and never select the provisioning workstation's internal drive.

9.6 Assign and record the hostname



The screenshot shows the 'Customisation: Choose hostname' screen in the Raspberry Pi Imager v2.0.11.1. On the left is a sidebar with menu items: Device, OS, Storage, Customisation (highlighted in red), Hostname (underlined), Localisation, User, Wi-Fi, Remote access, and Raspberry Pi Connect. The main area has the title 'Customisation: Choose hostname' and a text input field labeled 'Enter your hostname' containing 'MultimedicaScanner1'. Below the input field is a note: 'A hostname is a unique name that identifies your Raspberry Pi on the network. It should contain only letters, numbers, and hyphens.' At the bottom are four buttons: 'APP OPTIONS', 'SKIP CUSTOMISATION', 'BACK', and 'NEXT' (highlighted in red).

Enter a unique lowercase Pi hostname and click **NEXT**. Record it in the installation worksheet introduced in Section 3 because every later `-PiHost` command uses it.

Recommended examples:

```
multimedica-lab-01
multimedica-reg-01
```

Although hostnames are generally case-insensitive, lowercase avoids confusion when copying commands.

9.7 Configure localization

Raspberry Pi Imager v2.0.11.1

OS

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Customisation: Localisation

Select your location for suggested time zone and keyboard layout

Capital city: Washington, D.C. (United States) ⓘ

Time zone: America/Chicago

Keyboard layout: us

APP OPTIONS SKIP CUSTOMISATION BACK NEXT

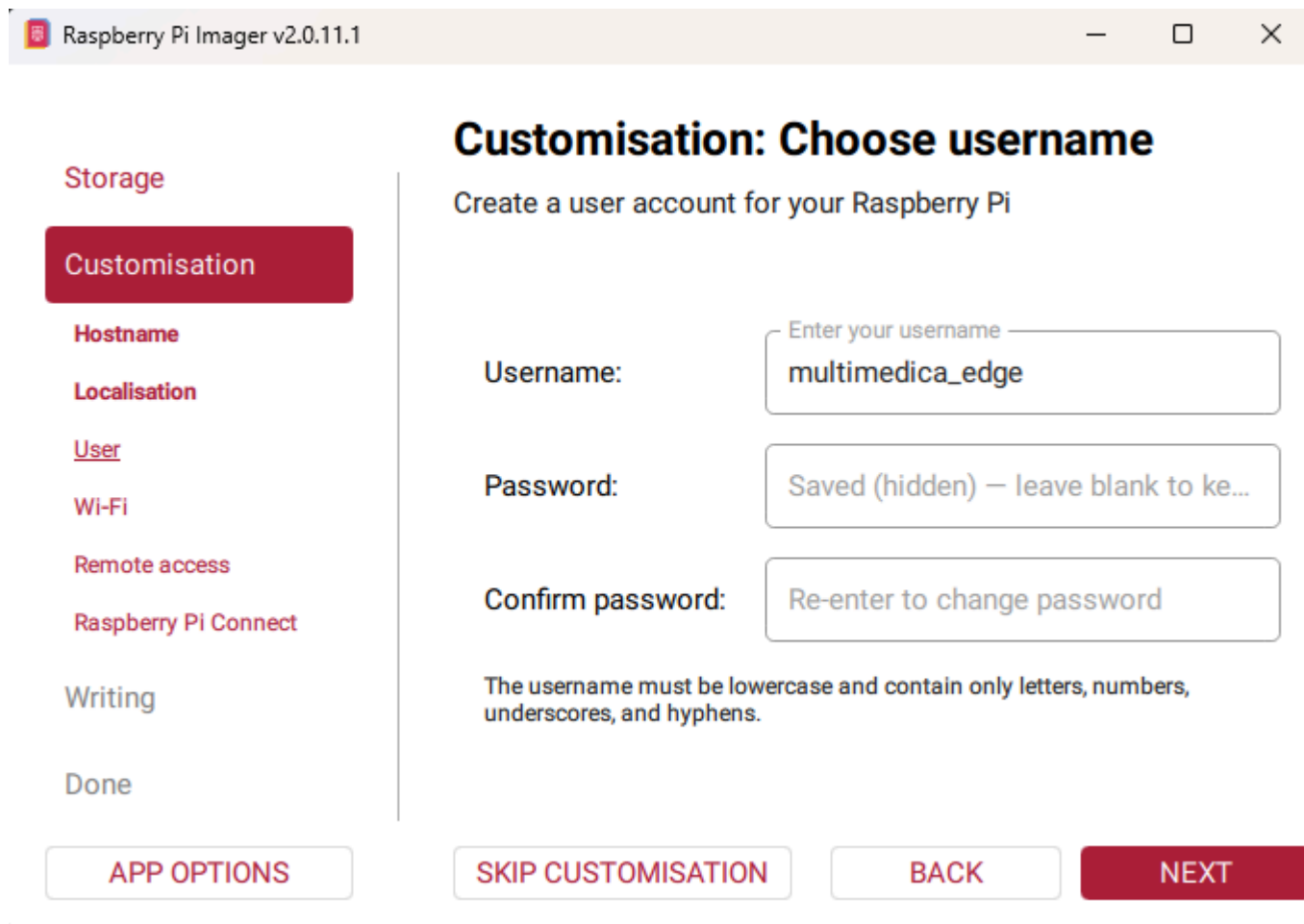
Configure localization for the appliance's operating location, not necessarily the provisioning workstation's location.

For a Wisconsin test appliance:

```
Capital city: Washington, D.C. (United States)
Time zone: America/Chicago
Keyboard layout: us
```

For an appliance deployed in Guatemala, select the corresponding Guatemala location and use [America/Guatemala](#). Use the `us` keyboard layout unless the support keyboard requires another layout. The time zone affects logs, displayed time, troubleshooting, and scheduled clinic behavior.

9.8 Create the Pi user



The screenshot shows the 'Customisation: Choose username' screen in the Raspberry Pi Imager v2.0.11.1. On the left is a sidebar with navigation options: Storage, Customisation (highlighted in red), Hostname, Localisation, User, Wi-Fi, Remote access, and Raspberry Pi Connect. Below these are 'Writing' and 'Done'. The main area is titled 'Customisation: Choose username' and 'Create a user account for your Raspberry Pi'. It contains three input fields: 'Username:' with the value 'multimedica_edge', 'Password:' with the text 'Saved (hidden) – leave blank to ke...', and 'Confirm password:' with the text 'Re-enter to change password'. A note states: 'The username must be lowercase and contain only letters, numbers, underscores, and hyphens.' At the bottom are four buttons: 'APP OPTIONS', 'SKIP CUSTOMISATION', 'BACK', and 'NEXT' (highlighted in red).

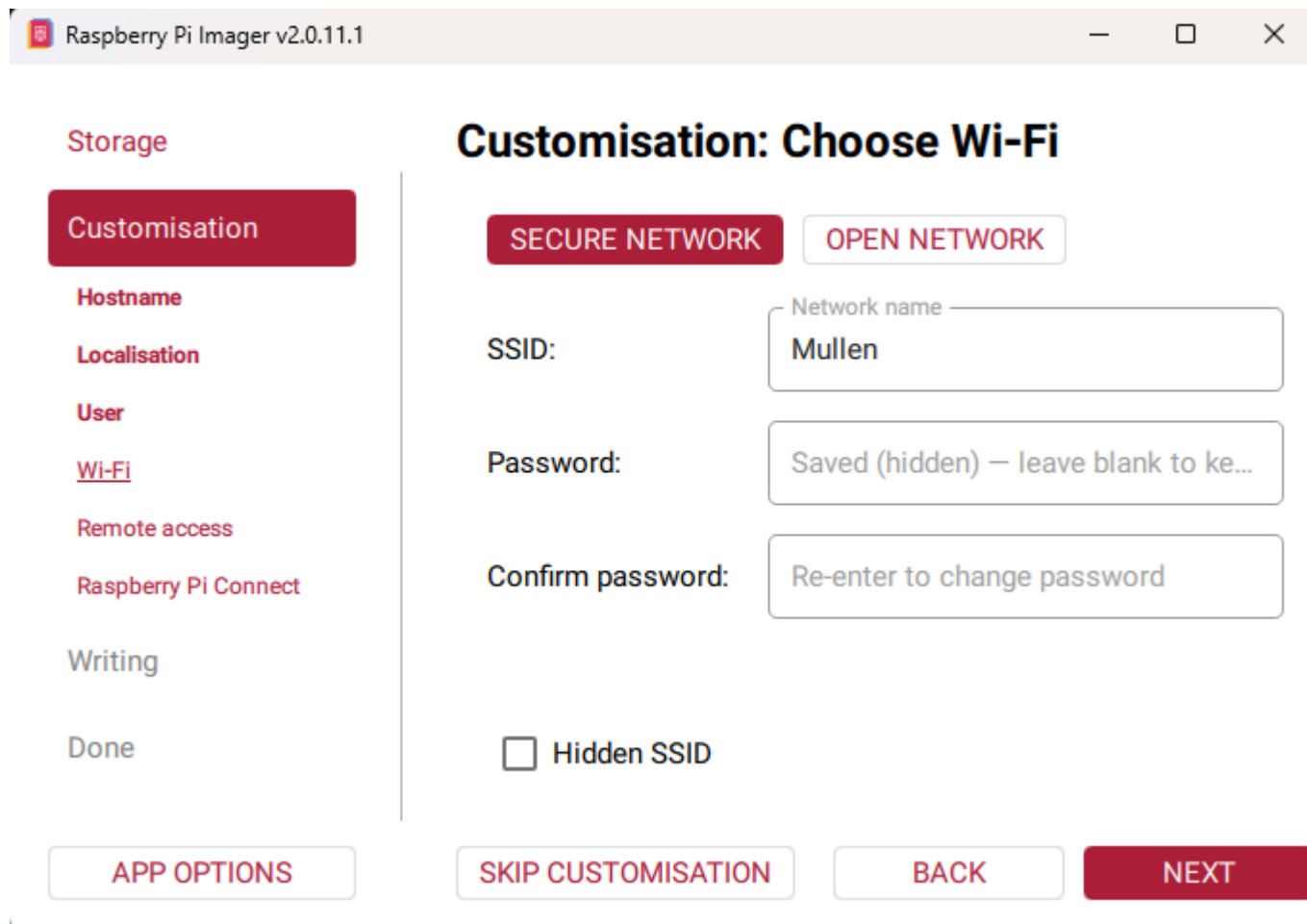
Enter the required username exactly:

```
multimedica_edge
```

The current Multimédica services and installer are configured to run under this specific Pi account. Unlike the Pi hostname, the username is not installer-selectable. Using a different username will cause installation or runtime checks to fail.

Create a strong unique password and enter it in both password fields. Store it in the approved password manager or secure credential record. Do not place it in the installation worksheet, Git, screenshots, or console transcripts.

9.9 Configure initial Wi-Fi access



Raspberry Pi Imager v2.0.11.1

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

Customisation: Choose Wi-Fi

SECURE NETWORK OPEN NETWORK

SSID: Network name Mullen

Password: Saved (hidden) – leave blank to ke...

Confirm password: Re-enter to change password

☐ Hidden SSID

APP OPTIONS SKIP CUSTOMISATION BACK NEXT

Image-time Wi-Fi is highly recommended because it lets the new Pi join the network immediately. Select **SECURE NETWORK**, enter the SSID and password, confirm the password, and click **NEXT**.

This step may be omitted when wired Ethernet will be connected before first boot and DHCP is available. Image-time Wi-Fi provides installation connectivity; it does not replace the later Multimedica Wi-Fi commissioning QR.

9.10 Enable SSH with password authentication

Raspberry Pi Imager v2.0.11.1

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

Customisation: SSH authentication

Configure SSH access

Enable SSH 

[Learn about SSH](#)

Authentication mechanism:

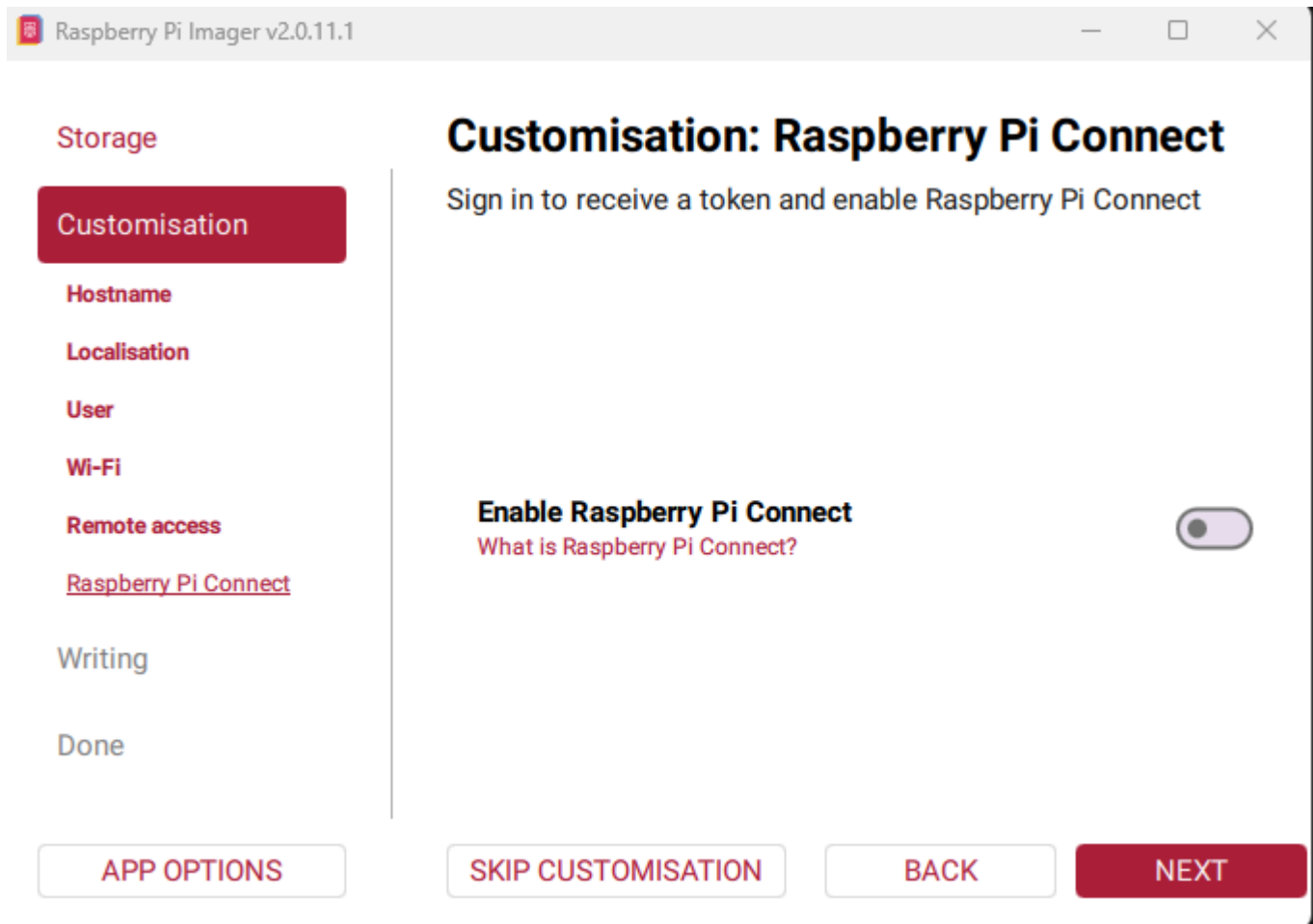
☒ Use password authentication

☐ Use public key authentication

APP OPTIONS SKIP CUSTOMISATION BACK NEXT

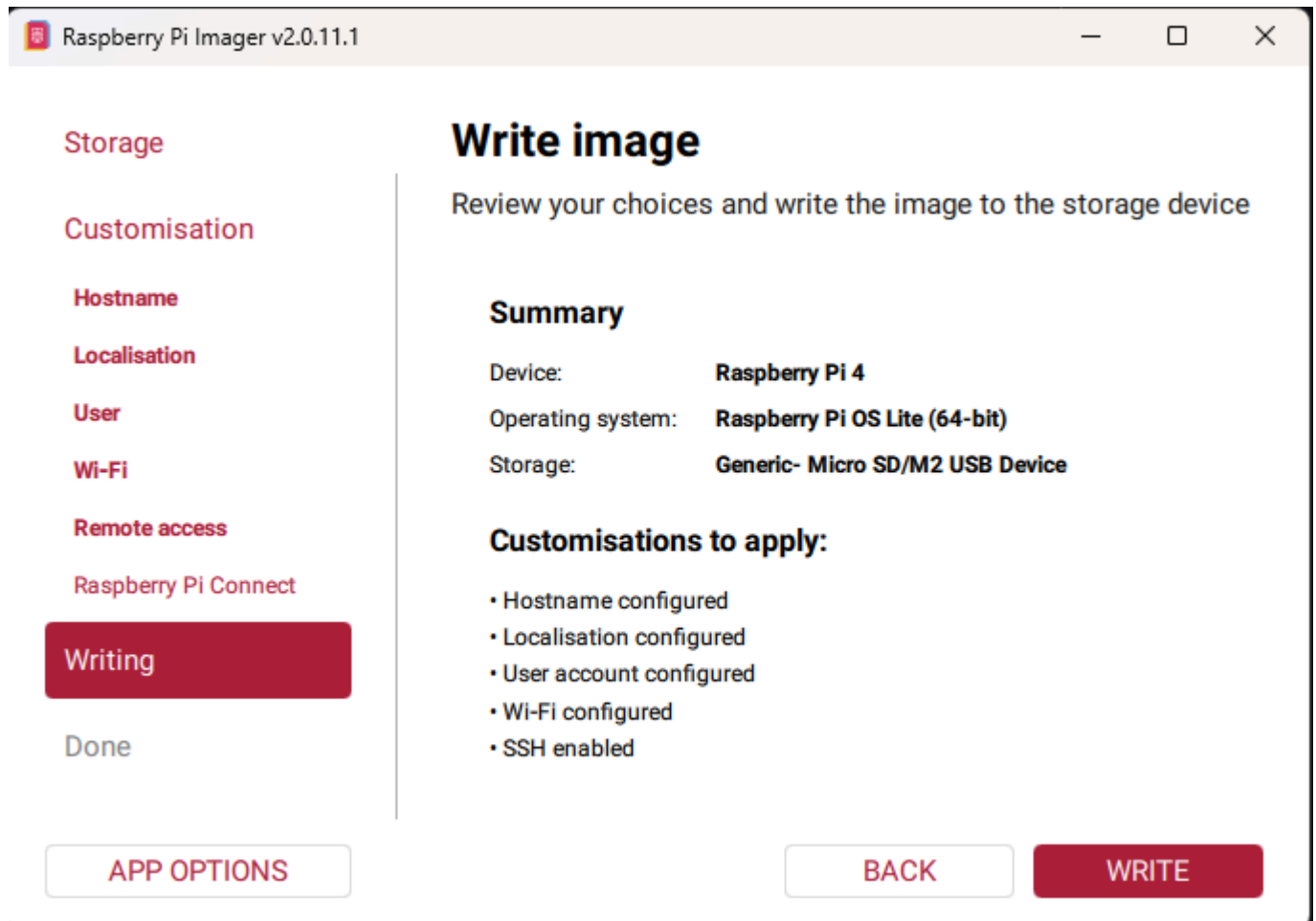
Turn on **Enable SSH**, select **Use password authentication**, and click **NEXT**. The later `-ConfigureSshAccess` operation installs and verifies the dedicated Multimedica provisioning key.

9.11 Leave Raspberry Pi Connect disabled



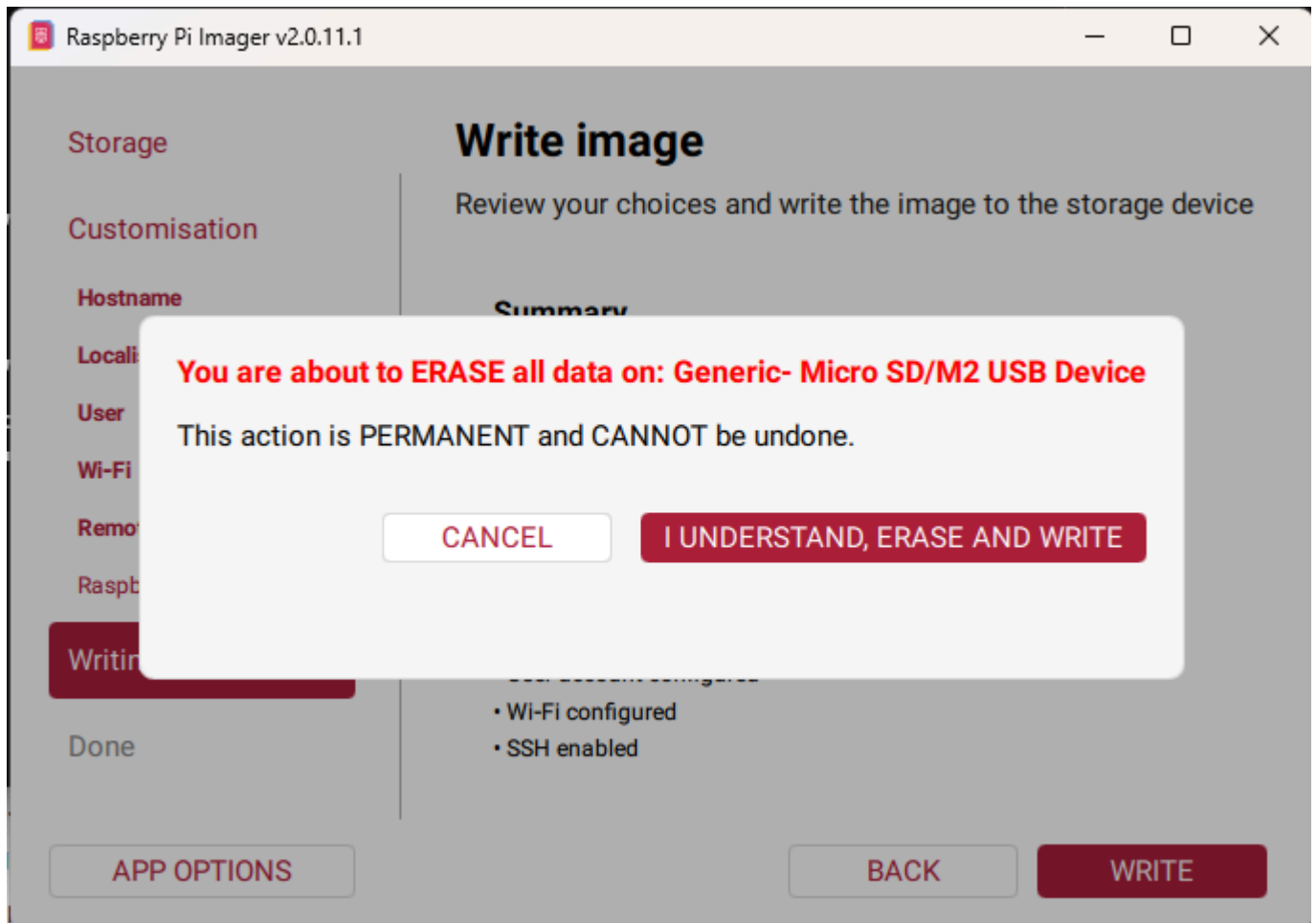
Leave **Enable Raspberry Pi Connect** turned off and click **NEXT**. The supported process uses direct SSH and does not require a Raspberry Pi Connect account or token.

9.12 Review and write



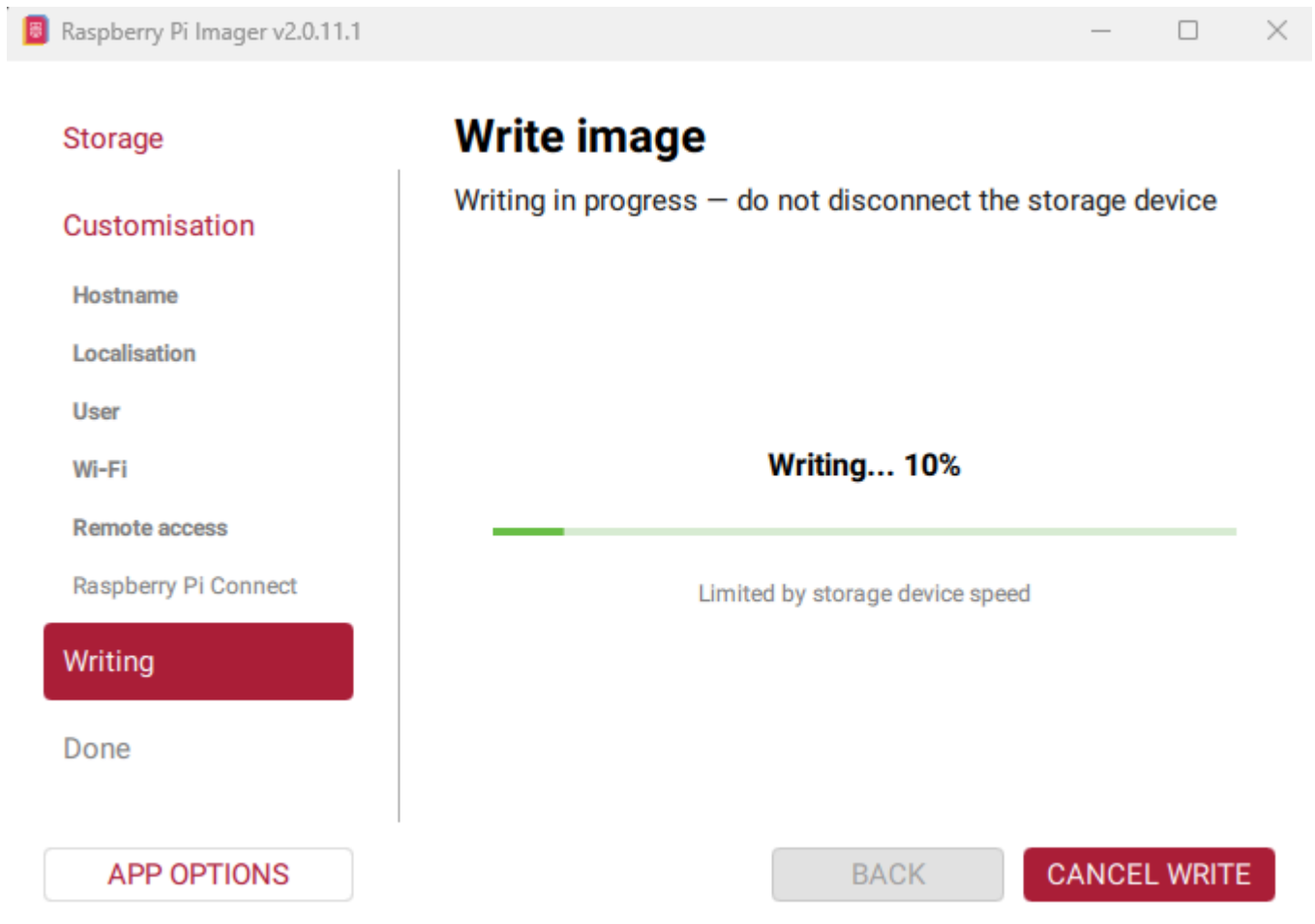
Confirm Raspberry Pi 4, Raspberry Pi OS Lite 64-bit, the intended microSD card, hostname, localization, user account, network choice, and SSH. If correct, click **WRITE**.

9.13 Confirm erasure



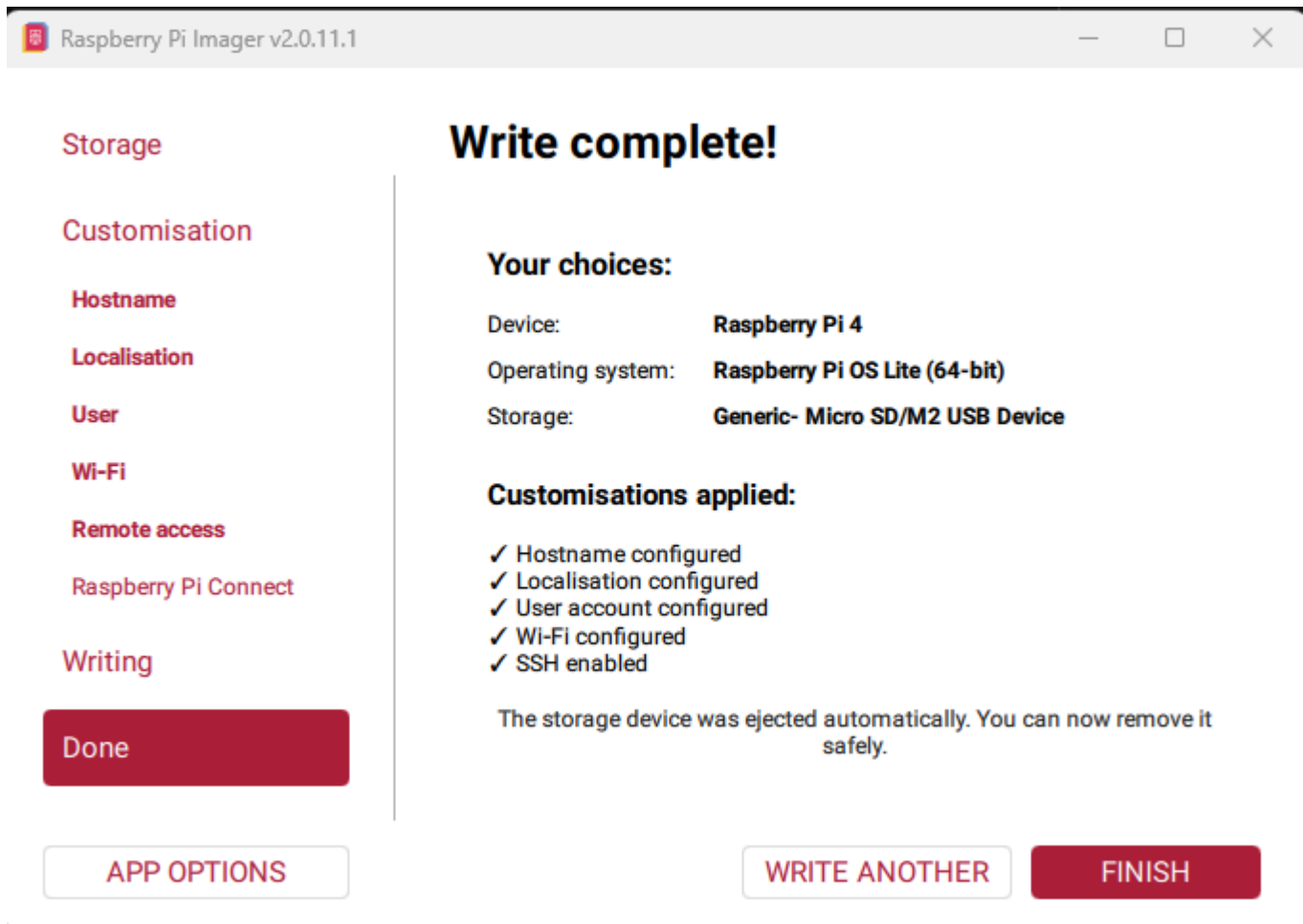
Verify the named device once more. Click **I UNDERSTAND, ERASE AND WRITE** only when it is positively identified as the intended microSD card. Otherwise click **CANCEL**.

9.14 Wait for writing and verification



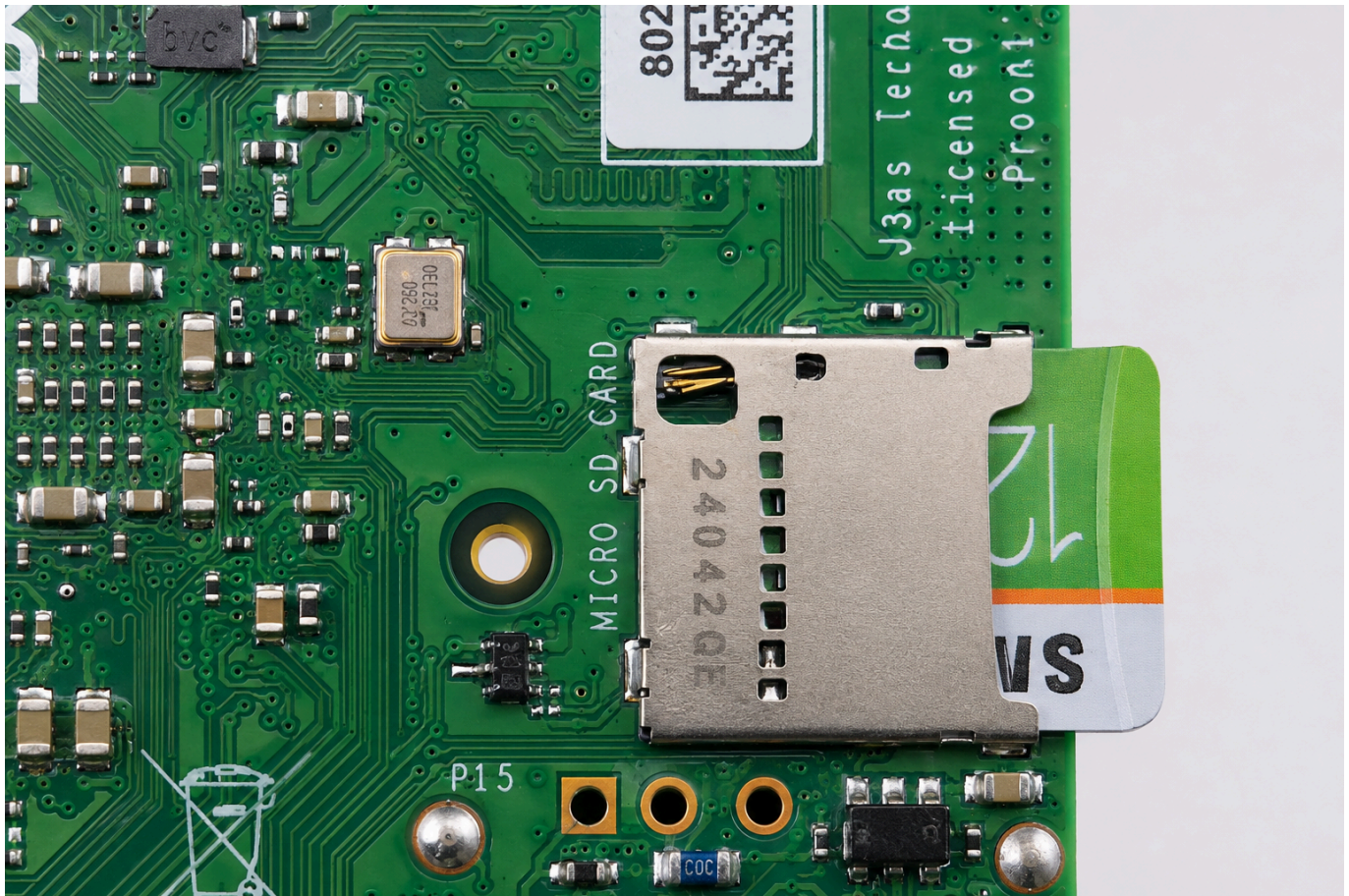
Allow both **Writing** and **Verifying** to complete. Do not cancel, remove the card, disconnect the reader, shut down the workstation, or allow it to sleep. Duration varies, and progress may pause temporarily without indicating failure.

9.15 Finish and remove the card



After Imager reports **Write complete!**, click **FINISH** and remove the automatically ejected microSD card.

Before first boot, insert the newly imaged microSD card into the slot on the underside of the Pi as shown. Push it fully into place without forcing or bending it.



Then place the electronics into the enclosure, route the external cables through their intended exits, dress and secure the cables, and confirm that no cable or connector is pinched or strained. Do not apply power until the enclosure assembly and cable check are complete.

9.16 First boot

1. Confirm that the microSD card is fully seated and the enclosure assembly is complete.
2. Connect the scanner, display, and network as applicable.
3. Apply power.
4. Allow the first boot to finish.

From **Windows PowerShell**, set the recorded values if they are not already present, then confirm that the Pi is reachable. Paste or enter all four lines in Windows PowerShell. After entering the final **Test-Connection** line, press **Enter** to run it:

```
$piUser = Read-Host "Enter the Pi username recorded during imaging"
$piHostname = Read-Host "Enter the Pi hostname recorded during imaging"
$piHost = "$piUser@$piHostname"
Test-Connection -ComputerName $piHostname -Count 2
```

If Windows cannot find the Pi using its hostname, obtain the Pi's IP address from the network router or DHCP system. Then run this replacement block in **Windows PowerShell**:

```
$piIp = Read-Host "Enter the Pi IP address"
$piHost = "$piUser@$piIp"
```

CHECKPOINT — Where you are now

The Pi has a clean operating system, its recorded network identity, and initial SSH access. Multimédica software has not yet been installed.

10. Establish provisioning SSH access

The installation software needs to control the new Raspberry Pi remotely so it can copy files and configure the Pi as a Multimédica scanner. **SSH** is the secure remote-control tool that allows the Windows provisioning workstation to run commands on the Pi. This step establishes and verifies that connection for the remaining installation work.

Windows PowerShell:

```
.\provision-scanner.ps1 `
  -ConfigureSshAccess `
  -PiHost $piHost
```

When asked whether the Pi was reimaged:

- answer lowercase **yes** for a newly imaged card;
- answer lowercase **no** only when the existing SSH identity is expected.

Enter the Pi account password when SSH requests it. The script installs and verifies a dedicated provisioning public key. Successful output ends with:

```
Key-based SSH verified; provisioning will not prompt for the Pi password
```

If SSH reports an identity change after answering **no**, stop and verify that this is the intended physical Pi.

11. Install the bootstrap platform

Windows PowerShell:

```
.\provision-scanner.ps1 `
  -Install `
  -PiHost $piHost `
  -InstallerConfig .\multimedica-installer.json `
  -ResultFile .\provisioning-result.json
```

At the sudo prompt, enter the Pi password once and press Enter. The password will not appear. Wait after pressing Enter; do not type it again unless another visible prompt appears.

The first installation typically takes **5–10 minutes** while operating-system and Node.js (**npm**) packages are installed. The Windows PowerShell window should continue to show installation progress. Short pauses are normal. If there is no new output for more than about a minute, first look carefully for a visible password or confirmation prompt; do not type blindly. If there is no prompt and the process remains unchanged, preserve the screen and use the troubleshooting guidance.

The script will:

- check the platform baseline;
- stage bootstrap files;
- install dependencies and services;
- install restricted privileged helpers;
- start and validate controller, display, and kiosk services;
- verify the scanner device and reader;
- reboot the Pi;
- verify that services recover after reboot.

Near the end, the script should report the reboot sequence. The hostname may appear in the actual output, but the progression should resemble:

```
Rebooting Pi (no operator input required)
The Pi will disconnect temporarily. Do not type a password or scan a QR code.
Waiting for the current system to go offline...
OK  Pi went offline
Waiting for the rebooted Pi to reconnect...
```

During installation, the physical screen may show:

```
Finalizando la instalación
Espere. No escanee códigos todavía.
```

Do not scan configuration QRs while this waiting message is displayed. Wait for the command to report **RESULT: PASS** and for the display to request configuration.

Required console result:

```
RESULT: PASS
```

CHECKPOINT — Where you are now

The operating system and Multimédica bootstrap layer are installed. The appliance can read commissioning QRs, but it is not yet configured for its clinic assignment and production software is not yet installed.

12. Commission the scanner appliance

You now have the bootstrap software installed—high five. **Commissioning** means telling the scanner about its particular clinic assignment: its Wi-Fi credentials, the type of station where it will operate (for example, laboratory or nursing), and how to reach the Multimédica cloud functions securely. Because the scanner does not have a keyboard, you provide this information using the QR codes prepared in Section 8.4.

Scan QRs only after bootstrap installation has passed and the display requests configuration.

Three QR codes are used to configure the appliance in this section: Wi-Fi, Station, and Cloud. A fourth QR, **Mostrar Identidad del scanner**, does not change configuration and is not required during these three commissioning scans. Keep it available for the Section 15.3 acceptance check and for later troubleshooting: it can display the scanner's identity, IP address, and installed version.

12.1 Wi-Fi QR

Scanner: Scan the Wi-Fi QR when Wi-Fi configuration is required.

The Pi applies and verifies the NetworkManager connection before storing it as authoritative. If the QR fails, do not continue on the assumption that image-time Wi-Fi is sufficient; preserve the message and diagnose the failure.

Applying different Wi-Fi credentials may temporarily interrupt SSH or change the Pi address.

12.2 Station QR

Scanner: Scan the Station QR.

Verify that the display acknowledges the station configuration in Spanish.

12.3 Cloud QR

Scanner: Scan the Cloud QR.

Verify that the display acknowledges the cloud configuration in Spanish.

When all required configuration is accepted, the display should show:

Configuración completa

The controller may retain previously valid Station or Cloud configuration on a reinstallation. If the display does not request one of those QRs, do not assume that a scan was ignored. Continue to Section 13 and run the complete **Windows PowerShell -Verify** command shown there. Its result reports whether the required configuration is actually complete.

13. Verify bootstrap and commissioning

Windows PowerShell:

```
.\provision-scanner.ps1 `
  -Verify `
  -PiHost $piHost `
  -ResultFile .\provisioning-result.json
```

Required observations include:

- configuration complete;
- display service active;
- controller service active;
- kiosk service active;
- display health endpoint responsive;
- physical Chromium kiosk active;
- scanner USB device detected;
- scanner reader active.

Required result:

```
RESULT: PASS
```

Do not install production if verification fails.

CHECKPOINT — Where you are now

You now have a basic, commissioned scanner appliance with verified scanner input, display, kiosk, network, station, and cloud configuration. The final installation step is to load and promote the approved production software.

14. Install the production release

Section 8.5 loaded and validated the repository's approved release. Confirm that those variables are still available in the current PowerShell window.

Windows PowerShell:

```
$releaseVersion
$artifact
$sha
```

All three commands must display nonblank values. If any is blank, return to section 8.5 and rerun its complete validation block. Then run the following in the same PowerShell window.

Windows PowerShell:

```
.\provision-scanner.ps1 `
  -InstallRelease `
  -PiHost $piHost `
  -ReleaseVersion $releaseVersion `
  -ArtifactPath $artifact `
  -ArtifactSha256 $sha `
  -ResultFile .\provisioning-result.json
```

The installer stages and validates the release, starts it as an isolated candidate, and asks:

Confirm the physical display showed the CANDIDATO state.
Type lowercase yes to continue:

Inspect the physical Pi display.

- If it visibly shows **CANDIDATO**, type lowercase **yes** and press Enter.
- If it does not, do not authorize promotion.

A successful installation ends with:

RESULT: PASS

The release is not recorded as known-good until production starts through the production service and passes health verification.

CHECKPOINT — Where you are now

The approved production release is installed and has passed candidate and first-activation checks. The display should now look as it will during ordinary clinic use. If the clinic is operating, the station should show its current daily state—for example, available or in use. Complete the physical and operational acceptance checks below before turning the appliance over for use.

15. Final appliance acceptance

15.1 Read-only verification

Run **-Verify** again after production installation.

Windows PowerShell:

```
.\provision-scanner.ps1 `
  -Verify `
  -PiHost $piHost `
  -ResultFile .\provisioning-result.json
```

Confirm **RESULT: PASS** and that production is active and healthy.

15.2 Physical clinic state

Confirm that the display:

- uses portrait orientation;
- fills the intended display area;
- uses the approved El Alfarero logo, colors, and layout;
- reports the correct station and clinic state;
- contains no unexpected English commissioning text.

15.3 Scanner identity QR

Scan the **Mostrar Identidad del scanner** QR.

Confirm that a temporary Spanish identity screen appears and shows:

- location ID;
- room ID;
- station ID;
- device ID;
- current IP address;
- installed production version.

Confirm that it contains no credentials and automatically returns to the clinic display after approximately 15 seconds.

15.4 Create and scan a test patient

Do not use a real patient for installation acceptance. In the main Multimédica application, create a temporary test patient using the normal clinic procedure. In **Motivo de visita**, enter exactly:

```
testing
```

Use lowercase letters as shown below. Complete the other required patient fields, select the station workflow being tested, and create the patient.

EL ALFARERO MULTIMÉDICA Paul's Office - 09:42 PM - 0 Pacientes Plan de atención de pacientes

Registro de paciente

* Nombre:

DPI:

* Edad: * Género:

Número de teléfono:

Organización:

* Motivo de visita:

* Tipo de Visita:

Enfermería	Lab: Resultados	Lab: Muestra
Urgencias	Obstetricia/Ginecología	Fisioterapia
Consulta Pediátrica	Consulta Adultos	Lectura externa
Lectura interna	Dentista	Nutrición
Psicología	Doctor visitante	Certificaciones
Referencias hospitalarias		

From the **Anfitrión** page, print the test patient's workflow ticket using the clinic's configured page-printer or thermal-printer process. Scan that printed test barcode with the scanner appliance.

Confirm:

- the scan is accepted by production;
- the expected cloud workflow occurs;
- the display updates to the expected clinic state;
- no obsolete "Patient Scan Accepted" overlay appears.

15.5 Remove the test patient

After the scan test is complete, remove the temporary test patient without affecting real patients:

1. In the Multimédica application, open the patient workflow screen.
2. Click **Plan de atención de pacientes** in the upper-right corner **five times in quick succession**.
3. Use the test-cleanup control that appears to clear patients whose **Motivo de visita** is exactly **testing**.
4. Confirm that the test patient is gone and that no real patient was removed.

This hidden cleanup method is deliberately limited to test patients identified by the lowercase **testing** visit reason.

15.6 Cold-boot recovery

Perform one controlled power cycle by unplugging the scanner appliance's power cable, waiting approximately 10 seconds, and plugging it back in. Allow the Pi to return without operator intervention.

Confirm:

- startup display remains stable rather than scrambled;
- bootstrap services return;
- release recovery authorizes the known-good production release;
- production becomes active;
- the correct clinic state returns.

Run `-Verify` one final time and retain the result with the installation record.

INSTALLATION COMPLETE

The scanner appliance is programmed, configured, tested, and ready for normal clinic use. Complete the installation record and completion checklist at the end of this document, then turn the appliance over to the clinic operator.

Stop here for an ordinary new-scanner installation. Section 16 is a separate reference procedure for future troubleshooting, display updates, software releases, and repair. It is not part of programming and commissioning a new Pi.

16. Separate procedure: future maintenance and updates

This section is information to keep available for a future service visit. Do not perform these operations merely because they appear after the installation steps. Use them only when diagnosing an existing appliance or deploying an approved update.

16.1 Diagnose without changing the Pi

Use the following command from the repository root.

Windows PowerShell:

```
.\provision-scanner.ps1 -Verify ...
```

This should be the first action for an uncertain appliance state.

16.2 Update approved display resources

This operation updates the supported display bundle without replacing production. Begin with a clean, current repository checkout as described in section 7.5.

Windows PowerShell — repository root:

```
.\provision-scanner.ps1 `
  -UpdateDisplay `
  -PiHost $piHost `
  -ResultFile .\provisioning-result.json
```

Enter the Pi sudo password when requested. Do not manually copy display files onto the Pi.

Required result:

```
RESULT: PASS
```

Afterward, run `-Verify`, inspect the physical display, and confirm the correct clinic state returns.

16.3 Install new production code

Obtain the approved new production artifact, version, and SHA-256 from the release owner. The authoritative repository is <https://github.com/paullmullen/multimedica-scanner>. Update its Windows checkout; do not update source code directly on the Pi.

Unless a qualified developer or release owner is actively supervising the work, the installer must deploy only code and artifacts that have already been approved and published through the normal release process. Do not edit source, invent a version, or build an unapproved production release during a field update.

From the repository root, run:

Windows PowerShell:

```
git status --short
git switch main
git pull --ff-only origin main
git status --short
git log -1 --oneline
```

Each `git status --short` command must produce no output before the next prompt. If either prints any status code or filename, stop and seek qualified help.

The updated repository must include its approved-release pointer, artifact, checksum, and build metadata. Rerun the complete validation block in section 8.5; it will establish `$releaseVersion`, `$artifact`, and `$sha` without asking the installer to select a version.

Windows PowerShell:

```
.\provision-scanner.ps1 `
  -InstallRelease `
  -PiHost $piHost `
  -ReleaseVersion $releaseVersion `
  -ArtifactPath $artifact `
  -ArtifactSha256 $sha `
  -ResultFile .\provisioning-result.json
```

Authorize promotion only when the physical display shows **CANDIDATO**. Require **RESULT: PASS**, then run **Verify** and repeat the relevant physical acceptance checks.

Do not overwrite an installed version directory and do not reuse a version for different artifact contents.

16.4 Publish an approved production release — release owner only

This subsection is not a field-installer procedure. A qualified developer or release owner builds, tests, and explicitly approves a new version before an ordinary installer receives it.

Build the new version.

Windows PowerShell — release owner only:

```
$newReleaseVersion = Read-Host "Enter the new semantic production version"
npm run build:production-release -- $newReleaseVersion .\release-output
```

Run the required release and regression tests. After technical and operational approval, publish the local pointer.

Windows PowerShell — release owner only:

```
node .\release\approve-production-release.js $newReleaseVersion
Get-Content .\release-output\approved-production-release.json
```

The approval command fails unless the artifact, checksum sidecar, and build metadata all exist and the artifact hash matches. Review the generated pointer, then stage the complete release set together.

Windows PowerShell — release owner only:

```
git add `
  ".\release-output\multimedica-production-$newReleaseVersion.tgz" `
  ".\release-output\multimedica-production-$newReleaseVersion.tgz.sha256" `
  ".\release-output\multimedica-production-$newReleaseVersion.build.json" `
  .\release-output\approved-production-release.json

git diff --cached --check
git status --short
```

Commit and push only after review. The approved pointer and all three referenced files must be present in the same approved repository revision. Never update the pointer to files that have not been committed and published.

16.5 Bootstrap repair

-Repair is intended for bootstrap states where release management permits broad repair. It intentionally refuses to overwrite a production-managed appliance when doing so could invalidate release and recovery state.

Do not bypass this refusal with manual root file copies. Escalate for a release-aware bootstrap upgrade or supported narrow update.

There is currently no general automatic bootstrap updater and the Pi does not run **git pull**. GitHub is the source for the Windows provisioning workstation; validated files and production artifacts reach the Pi only through supported provisioning operations.

17. Installation record

Record at minimum:

```
Installation date:
Installer:
Hostname:
Pi IP address:
Location:
Station:
Room ID:
Device ID:
Qualified image:
Repository revision:
Production version:
Production SHA-256:
Initial Verify result:
Post-release Verify result:
Cold-boot Verify result:
Physical display accepted:
Real scan accepted:
Notes:
```

Do not record passwords, QR administrator tokens, shared secrets, or Wi-Fi credentials in the installation record.

18. Completion checklist

- ☐ Qualified hardware assembled correctly
- ☐ Raspberry Pi Imager completed writing and verification
- ☐ Pi imaged with correct user and SSH settings
- ☐ **-ConfigureSshAccess** completed
- ☐ **-Install** returned **RESULT: PASS**
- ☐ **-Install** passed and the display requested configuration before QR scanning
- ☐ Required QRs accepted
- ☐ Display reported **Configuración completa**

- ☐ Pre-release -Verify returned RESULT: PASS
- ☐ Production artifact hash matched
- ☐ Physical display showed CANDIDATO
- ☐ -InstallRelease returned RESULT: PASS
- ☐ Post-release -Verify returned RESULT: PASS
- ☐ Correct clinic state displayed
- ☐ **Mostrar Identidad del scanner** QR showed the correct IDs, IP address, and production version
- ☐ Real patient scan accepted
- ☐ Cold boot recovered production and stable display
- ☐ Final -Verify returned RESULT: PASS
- ☐ Installation record completed without secrets